

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) — Semester VI Examination, 2022-23

Paper No. MTB-605: Vector and Tensor Analysis

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — 2 marks each.)

- (a) If $\phi(x, y, z) = 2yz^2$ and $\vec{A} = xy\vec{i} + y^2z\vec{j} + xz\vec{k}$, obtain $\text{div}(\phi\vec{A})$. [2]
- (b) Given $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, show that $\nabla(r^n) = nr^{n-2}\vec{r}$. [2]
- (c) Give the definition of divergence $\nabla \cdot \vec{V}$ of a smooth vector field $\vec{V}(x, y, z) = V_1\vec{i} + V_2\vec{j} + V_3\vec{k}$. Obtain $\nabla \cdot \vec{V}$ for $\vec{V} = yz\vec{i} + zx\vec{j} + xy\vec{k}$. [2]
- (d) Give the definition of curl $\nabla \times \vec{V}$ of a smooth vector field $\vec{V}(x, y, z) = V_1\vec{i} + V_2\vec{j} + V_3\vec{k}$. Obtain $\nabla \times \vec{V}$ for $\vec{V} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$. [2]
- (e) Given $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, prove that $\nabla^2(1/r) = 0$ for $r \neq 0$. [2]
- (f) Give the definitions of conservative and irrotational vector fields. [2]
- (g) State Gauss' divergence theorem and Green's theorem in a plane. [2]
- (h) Define an orthogonal curvilinear coordinate system in $\mathbb{R}^3$. [2]
- (i) Write down the coordinate surfaces and coordinate curves for the spherical coordinate system in $\mathbb{R}^3$. [2]
- (j) Obtain the element of arc length for the cylindrical coordinate system in $\mathbb{R}^3$. [2]
- (k) Give the definition of contravariant and covariant vectors. [2]

Question 2.

- (a) Prove that $\nabla \times (\nabla\phi) = 0$ for any smooth scalar field ϕ . [6]
- (b) Prove that $\nabla \cdot (\nabla \times \vec{V}) = 0$ for any smooth vector field $\vec{V}$. [6]

Question 3.

- (a) If $\vec{V} = 2y^3\vec{i} + yz^2\vec{j} + zx^2\vec{k}$, evaluate $\int_C \vec{V} \cdot d\vec{r}$ along the path $x = t, y = t^2, z = t^3$ from $t = 0$ to $t = 1$. [6]

- (b) If $\int_C \vec{V} \cdot d\vec{r}$ is independent of the path C joining any two points, show that there exists a scalar function ϕ such that $\vec{V} = \nabla\phi$. [6]

Question 4.

- (a) If $\vec{V} = y\vec{i} + (x - 2xz)\vec{j} - xy\vec{k}$, evaluate $\iint_S (\nabla \times \vec{V}) \cdot \vec{n} dS$, where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ above the xy -plane. [6]
- (b) Using Green's theorem, show that the area bounded by a simple closed curve C is $\frac{1}{2} \oint_C (x dy - y dx)$. Hence obtain the area enclosed by the ellipse $x = 2 \cos \theta$, $y = 3 \sin \theta$. [3+3]

Question 5.

- (a) State and prove Gauss' divergence theorem and show that for a closed surface S :

$$\iiint_S \vec{r} \cdot \vec{n} dS = 3V$$

where V is the volume enclosed by S . [6]

- (b) Verify Stokes' theorem for $\vec{V} = (2x - y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$, where S is the upper hemisphere of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary. [3+3]

Question 6.

- (a) Evaluate the triple integral:

$$\iiint_V (x^2 + y^2 + z^2) dx dy dz$$

where V is the region enclosed by the sphere $x^2 + y^2 + z^2 = a^2$. [6]

- (b) Express $\nabla \times \vec{V}$ and $\nabla^2\phi$ in spherical polar coordinates in $\mathbb{R}^3$. [3+3]

Question 7.

- (a) Give the definition of Christoffel symbols of the second kind and obtain the transformation laws for them. [2+4]
- (b) Prove that the covariant derivatives of g_{ij} , g^{ij} , and δ_j^i are zero. [2+2+2]

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